down the document into smaller sections, summarizing each chunk independently, and then aggregating these summaries into an overview. This multistage process is particularly effective for handling extensive texts, as it allows for parallelization of tasks. Figure 7.2 shows this approach close up.

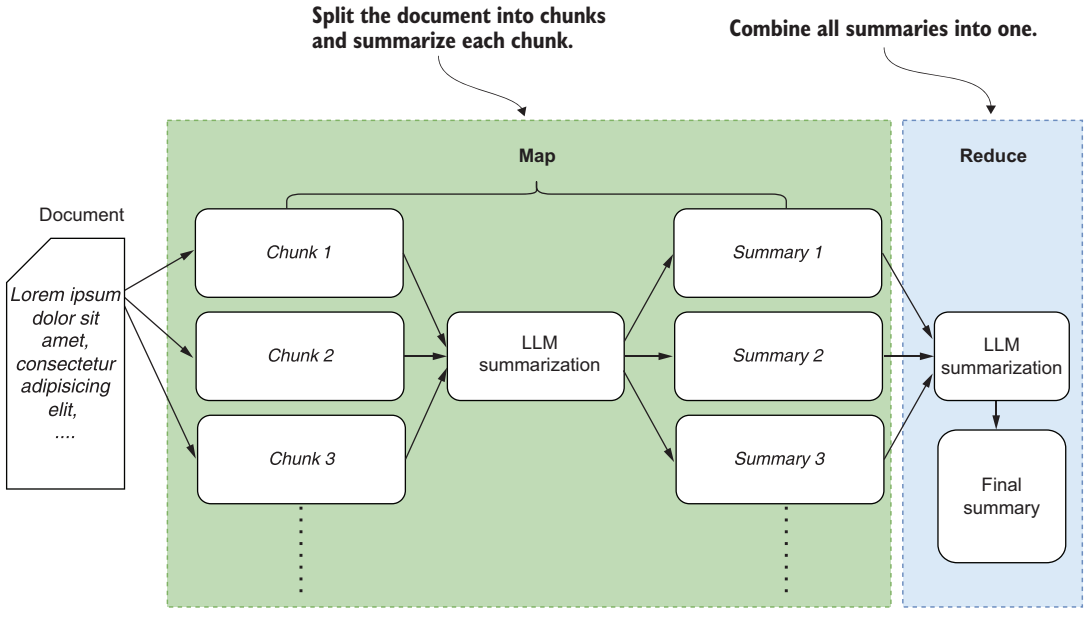


Figure 7.2 The MapReduce summarization pattern, behind the scenes. The input document is divided into smaller chunks. Each chunk is then processed independently by the map function, generating intermediate summary pairs. These pairs are subsequently organized and are grouped together. The reduce step is then applied to these grouped pairs, aggregating the values to produce a final summary.

Here's a breakdown of the steps:

1. *Split document into chunks*—The first step is to split the original document into smaller "chunks" or sections.
2. *Summarize each chunk*—For each chunk, a summarization process is applied to generate a summary of the content in that chunk.
3. *Map*—The summaries of each chunk are then "mapped" to an intermediate representation inside a data structure that keeps track of the individual summaries.
4. *LLM summarization*—The intermediate representation of the individual chunk summaries is then passed through an LLM summarization process. This step takes the individual chunk summaries and combines them into a high-level summary.
5. *Reduce*—The final step is to "reduce" the LLM-generated summaries into a single, combined summary that represents the overall document.